

The Independence Axiom in Expected Utility Theory

ECO310 Midterm Review

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1 Introduction to Expected Utility

Expected Utility Theory provides a framework for understanding decision-making under uncertainty. For preferences over lotteries to be represented by expected utility, they must satisfy four key conditions:

1. **Completeness:** For any two lotteries p and q , either $p \succeq q$ or $q \succeq p$ (or both)
2. **Transitivity:** If $p \succeq q$ and $q \succeq r$, then $p \succeq r$
3. **Continuity:** Small changes in probabilities lead to small changes in preferences
4. **Independence Axiom:** The critical condition for expected utility

2 The Independence Axiom

2.1 Formal Statement

The independence axiom states that if you prefer lottery p to lottery q , then you should still prefer p to q even when both are mixed with some third lottery r in the same proportion.

Independence Axiom

If $p \succ q$, then for any lottery r and probability $\alpha \in (0, 1)$:

$$\alpha p + (1 - \alpha)r \succ \alpha q + (1 - \alpha)r$$

2.2 Intuition

The independence axiom captures the idea that your preference between two lotteries should not depend on what happens in states that are common to both. If you prefer p to q , mixing both with the same lottery r shouldn't reverse your preference.

Think of it this way: if you prefer apples to oranges, then you should still prefer "80% chance of apples, 20% chance of bananas" to "80% chance of oranges, 20% chance of bananas."

3 Sally's Example: Violating Independence

3.1 The Setup

Consider three outcomes with monetary values:

- $x_1 = \$1,000,000$
- $x_2 = \$5,000,000$
- $x_3 = \$0$

Sally expresses the following preferences over lotteries (where each lottery is written as $(P(x_1), P(x_2), P(x_3))$):

$$\text{Preference 1: } (0, 1, 0) \succ (0.01, 0.89, 0.10) \tag{1}$$

$$\text{Preference 2: } (0.90, 0, 0.10) \succ (0.89, 0.11, 0) \tag{2}$$

3.2 Interpreting the Preferences

Preference 1: Sally prefers a guaranteed \$5,000,000 to a lottery with:

- 1% chance of \$1,000,000
- 89% chance of \$5,000,000
- 10% chance of \$0

This shows Sally is somewhat risk-averse—she'd rather have the sure thing than take a 10% chance of getting nothing.

Preference 2: Sally prefers a lottery with 90% chance of \$1,000,000 and 10% chance of \$0 to a lottery with 89% chance of \$1,000,000 and 11% chance of \$5,000,000.

This seems to show Sally is willing to give up a potential extra \$4,000,000 to increase her chance of winning something from 89% to 90%.

3.3 The Key Insight

Here's the crucial observation: **the two left-hand-side lotteries add up to the same total as the two right-hand-side lotteries.**

Let's verify:

$$\begin{aligned} (0, 1, 0) + (0.90, 0, 0.10) &= (0.90, 1.00, 0.10) \\ (0.01, 0.89, 0.10) + (0.89, 0.11, 0) &= (0.90, 1.00, 0.10) \end{aligned}$$

They're identical!

4 Proving the Violation

4.1 What Expected Utility Would Require

If Sally's preferences could be represented by expected utility, there would exist a utility function u such that preferences are determined by expected utility values.

From Preference 1:

$$\begin{aligned}u(x_2) &> 0.01u(x_1) + 0.89u(x_2) + 0.10u(x_3) \\u(x_2) - 0.89u(x_2) &> 0.01u(x_1) + 0.10u(x_3) \\0.11u(x_2) &> 0.01u(x_1) + 0.10u(x_3)\end{aligned}$$

From Preference 2:

$$\begin{aligned}0.90u(x_1) + 0.10u(x_3) &> 0.89u(x_1) + 0.11u(x_2) \\0.90u(x_1) - 0.89u(x_1) + 0.10u(x_3) &> 0.11u(x_2) \\0.01u(x_1) + 0.10u(x_3) &> 0.11u(x_2)\end{aligned}$$

4.2 The Contradiction

We have derived:

$$\text{From Preference 1: } 0.11u(x_2) > 0.01u(x_1) + 0.10u(x_3)$$

$$\text{From Preference 2: } 0.01u(x_1) + 0.10u(x_3) > 0.11u(x_2)$$

Contradiction!

These two inequalities cannot both be true! We have:

$$0.11u(x_2) > 0.01u(x_1) + 0.10u(x_3) > 0.11u(x_2)$$

which implies $0.11u(x_2) > 0.11u(x_2)$, a logical impossibility.

4.3 Why Independence Fails

The independence axiom would say: if we can write Preference 1 as a mixture of two lotteries where one lottery appears in Preference 2, then the preferences should be consistent.

Notice that we can decompose the lotteries as:

$$\begin{aligned}(0, 1, 0) &= 0.11 \times (0, 1, 0) + 0.89 \times (0, 1, 0) \\(0.01, 0.89, 0.10) &= 0.11 \times (0, 0, 1) + 0.89 \times (0, 1, 0) + 0.01 \times (1, 0, 0)\end{aligned}$$

The mixing structure reveals the inconsistency when combined with Preference 2.

5 Implications

5.1 Real-World Behavior

Sally's preferences are an example of the famous **Allais Paradox**, which demonstrates that real people often violate the independence axiom. This suggests:

- Expected utility theory may not perfectly describe actual human decision-making
- People treat certainty (100% probability) differently than near-certainty (89% or 90%)
- Alternative models like Prospect Theory have been developed to account for such violations

5.2 Why Independence Matters

The independence axiom is crucial because:

1. It's necessary (along with the other axioms) for preferences to be representable by expected utility
2. It ensures logical consistency in decision-making
3. Violations lead to predictable patterns of behavior that can be exploited
4. It separates rational choice theory from descriptive theories of actual behavior

6 Summary

- The **independence axiom** states that preferences between lotteries should not change when both are mixed with a third lottery in the same proportion
- Sally's preferences show: $(0, 1, 0) \succ (0.01, 0.89, 0.10)$ and $(0.90, 0, 0.10) \succ (0.89, 0.11, 0)$
- These preferences lead to a **mathematical contradiction** when we try to represent them with expected utility
- The contradiction proves Sally violates the independence axiom
- Such violations are common in real-world decision-making and motivate alternative theories